

Assumptions

- **Doctor/ Patient Onboarding and participation:** We assume that doctors and patients can be registered in the system before their first appointment is scheduled. Also, we assumed that one lead doctor is recorded as responsible for an appointment.
- **Equipment Need:** We assumed that certain services (like basic consultations) do not require medical equipment. This makes the participation between Medical_Service and Equipment partial.
- We interpreted the quotes “Each doctor can treat multiple patients and provide multiple services.” And “Patients can book appointments with multiple doctors and receive multiple services over time.” To mean that each appointment is associated with one doctor responsible for the appointment/ visit since the second quote does not explicitly say the patients can book an appointment with multiple doctors.

Alternative Design Choices

- **Doctor_Service Table.** We considered a “Doctor_Service” entity to define services rendered by each doctor.
 - We rejected this idea to keep the design in alignment with the requirement document. Interpreting “Doctors can provide multiple types of services” as saying each doctor can provide all services or doctors will be appointed to patients based on services to be provided.
- **Junction Table for Service Requirements.** We considered putting EquipmentID directly in Medical_Service.
 - We rejected this because a single service often requires multiple devices (e.g., an Ultrasound machine and Fetal monitor (CTG) can be used for a pregnancy scan service). A many-to-many junction (Service_Equipment) is more scalable for complex medical procedures.